

Md Rayhanul Masud

PhD in Computer Science, University of California, Riverside

Riverside, CA — Open to Postdoctoral and ML/AI Industry Positions

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Summary

PhD in Computer Science from UC Riverside with two years of software engineering experience. Research interests span machine learning, large language models, reinforcement learning, alignment, post-training, and model evaluation. Experienced in developing ML-driven systems and analyzing large-scale real-world datasets. Based in Riverside, California, and open to postdoctoral and ML/AI industry positions.

Education

University of California, Riverside

Ph.D. in Computer Science

Sep 2021 – Jun 2026

Advisor: Prof. Michalis Faloutsos

University of California, Riverside

M.S. in Computer Science

Sep 2021 – Dec 2024

CGPA: 3.90/4.00

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

Apr 2012 – Feb 2017

Advisor: Prof. Md Monirul Islam

Research & Engineering Experience

Graduate Student Researcher

Sep 2021 – Jun 2026

University of California, Riverside

- Designed and implemented RepoScope, an ML-powered “search by example” system for GitHub repositories, combining interpretable metadata features with code embeddings and visual clustering to enable multi-granularity exploration of large code corpora. **RepoScope** improved Top-5 precision by 20% vs baselines in our evaluation.
- Developed **GeekMAN** to match stylized usernames across security forums, GitHub, and social networks using string-similarity features, achieving up to 86% precision in our evaluation.
- Developed **MAGNET** to characterize software engineering practices and development patterns in malicious GitHub repositories.
- Built reproducible ETL pipelines for GitHub + forum data, defining feature schemas/mapping logic and using SQL/PySpark + Python to validate joins and produce analysis-ready tables and model inputs.

Software Engineer

Jul 2018 – Jul 2020

Kona Software Lab Ltd., Bangladesh

- Developed the transaction management pipeline for the **Kona Blockchain Platform** built on Ethereum; implemented a template-based smart-contract deployment mechanism that reduced insecure transaction requests from client applications by incorporating authorized smart contract interaction.
- Collaborated in an agile, product-focused team to prototype scalable blockchain solutions, contributing to design, implementation, testing, and deployment.

Publications

Published & Accepted Papers

- M. R. Masud**, A. T. Wasi, M. R. Parvez. “Reward Engineering for Software Tasks: A Survey of Reinforcement Learning Approaches.” *EMNLP (Findings)*, 2026. [\[Link\]](#)
- M. R. Masud**, Q. Zhang, M. Faloutsos. “MAGNET: A Software-Engineering View of the MAlware Ecosystem on GitHub NETwork.” *ASONAM*, 2026.

- **M. R. Masud**, M. O. F. Rokon, Q. Zhang, M. Faloutsos. “MetaSim: A search engine for finding similar GitHub repositories.” *International Conference on Software Maintenance and Evolution (ICSME)*, 2024. [\[Link\]](#) [\[Demo\]](#) [\[Platform\]](#)
- **M. R. Masud**, B. Treves, M. Faloutsos. “Disambiguating usernames across platforms: the GeekMAN approach.” *Social Network Analysis and Mining (SNAM)*, 2024. [\[Link\]](#)
- N. A. Tania, **M. R. Masud**, M. O. F. Rokon, Q. Zhang, M. Faloutsos. “Who is Creating Malware Repositories on GitHub and Why?” *The Web Conference (WWW)*, 2024. [\[Link\]](#)
- **M. R. Masud**, B. Treves, M. Faloutsos. “GeekMAN: Geek-oriented username Matching Across online Networks.” *ASONAM*, 2023. [\[Link\]](#) [\[Presentation\]](#)

Under Review

- **M. R. Masud**, M. R. Parvez. “TAME: Token Attribution and Masking for Emergent Misalignment.”
- **M. R. Masud**. “Equal Tokens, Unequal Documents: Long-Context Question Answering in Bangla.”
- **M. R. Masud**, Q. Zhang, M. Faloutsos. “RepoScope: A Multi-Granularity Query-by-Example Framework for GitHub Repository Search.”

Technical Skills

Languages: Python, Java, JavaScript, C++

ML / LLM: PyTorch, Transformers, Hugging Face

Data / NLP: Pandas, PySpark, scikit-learn, NLTK

Databases: SQL (MySQL), NoSQL

Systems / Tools: Docker, Git, Linux

Cloud: Google Cloud, AWS

Selected ML & AI Projects

Adversarial Prompt Generation for LLMs

[\[PDF\]](#)

- Employed a Morse-code-embedded adversarial prompt generation approach to evaluate the safety alignment robustness of LLaMA-2-7B-chat, achieving an 82.1% success rate on the AdvBench dataset.
- Implemented the full experimental pipeline for data generation, prompt engineering, and evaluation.

CalcGPT: Evaluating LLMs on Arithmetic

[\[Code\]](#)

- Implemented an evaluation framework to compare GPT-Neo-2.7B and LLaMA-7B on arithmetic tasks.
- LLaMA-7B achieved 86.1% accuracy on addition tasks, outperforming GPT-Neo-2.7B in our evaluation.

Search Engine for Twitter Data

[\[PDF\]](#)

- Designed and implemented a Twitter search engine on 2.5GB of data, benchmarking Hadoop vs. Lucene retrieval. Lucene built inverted indexes in $\sim 10.7\text{--}12.0\text{s}$ vs. Hadoop $38.6\text{--}140.0\text{s}$ (10k–50k tweets), while evaluating indexing efficiency, query latency, and relevance across configurations.

Service & Awards

Service

- External Reviewer: ACL ARR 2025; Judge: Cutie Hack 2025; Student Volunteer: KDD 2023.
- Community Contributor: Ethereum Stack Exchange (500+ reputation, $\sim 32\text{K}$ users reached).

Awards

- Dissertation Completion Fellowship Award, University of California, Riverside, 2026.
- Dean’s Fellowship Award, UC Riverside, 2021
- Dean’s Award (Level-4, Term-1), BUET, 2016.